

WardSync — What the Backend Can Do

A quick overview for the team. No technical jargon — just what exists and who can use it.

The 3 Roles

Nurse

- Registers new patients (fills in wristband number, triage color, photo, sex, age range)
- Records and edits vital signs
- Searches and views patient list
- Views room capacity and medication list

Doctor

- Views patients in their assigned room (can request to see all rooms)
- Changes patient status (waiting → in treatment → discharged / deceased)
- Records and edits both vital signs and treatments
- Selects medications from inventory when creating a treatment

Admin

- Creates and manages staff accounts (nurse / doctor / admin)
 - Manages medication inventory (add, edit, restock, delete)
 - Sets maximum bed count for each room
 - Views statistics and charts
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Features

1. Login & Profile Setup

Staff log in with email and password. First-time users must fill in their name, role, and assigned room before they can use the app. Doctors and admins must have an assigned room (Red / Yellow / Green / Black). Nurses do not need one.

2. Patient Registration

Nurses register patients at the triage point. Each patient gets a wristband number, triage color, sex, age range, and optionally a photo. Once registered, doctors in the same room receive a push notification immediately.

3. Patient List & Search

Anyone can view the patient list. Filters available:

- By room, status, or triage color
- By wristband number (type part of it)
- Only today's patients
- Custom time range (e.g. patients admitted between 8am–5pm)

All filters can be combined.

4. Push Notifications

The app sends automatic notifications — no manual trigger needed:

Situation	Who gets notified
New patient registered	Doctors in that room
Vital signs reach critical level	Doctors in that room
Medication stock falls low	All admins
Medication runs out	All admins

For notifications to work, the Flutter app must save the device's FCM token to the server after login.

5. Vital Signs

Nurses and doctors can record vital signs for a patient. The system automatically checks if any value is dangerous:

Measurement	Safe Range
Blood Pressure	90–180 systolic / 60–120 diastolic
Heart Rate	40–150 bpm
Temperature	35–39.5 °C
SpO2	90% and above
Respiratory Rate	8–30 per minute

If a value is outside the safe range, the app should highlight it in red, and the doctor receives a push notification.

Vital signs can be added, edited, and deleted individually (not just the whole record).

6. Treatments

Doctors record diagnosis, treatment plan, and notes. They can also attach medications from the inventory — when they do, stock is automatically reduced by one per medication added.

Treatments can be edited and deleted.

7. Admin Dashboard

Admins can see a summary of the whole hospital:

- How many patients total, broken down by triage color, room, and status
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8. Medication Inventory

Admins manage the medication list. Each medication has a name, default dosage, current stock, and a low-stock warning level.

- When stock drops to or below the warning level → admins get a push notification
- When stock hits zero → medication is marked Out of Stock and cannot be prescribed
- Admins can restock by adding or removing any amount

Nurses and doctors can browse the medication list (for reference or to select when writing a treatment).

9. Room Capacity

Each room (Red / Yellow / Green / Black) has a maximum bed count set by the admin. The system shows how full each room currently is, as a number and a percentage.

Color indicators:

- **Normal** — below 80%
- **Yellow warning** — 80% or more
- **Red warning** — 95% or more

Capacity updates automatically as patients are admitted or discharged.

10. Search & Filter

Already covered in Feature 3 above — all filters work together. Clearing filters just means sending no filter at all.

11. Statistics Charts (Admin only)

The admin dashboard can show charts for any time period (default: last 7 days):

- **Patients per day** — bar chart
 - **Patients by triage color** — pie chart
 - **Daily mortality rate** — line graph
 - **Average response time** — how long from patient arrival to starting treatment (in minutes)
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How It All Connects (for the Flutter team)

1. **Login** → get a token → store it → attach it to every request
2. **On app start** → check if profile is complete → if not, show setup screen
3. **After login** → save FCM token to the server (so notifications work)
4. **Before the treatment form** → load the medication list so the doctor can pick from it
5. **When showing vitals** → check the alerts field → highlight anything critical in red
6. **Room capacity screen** → use the warning field to decide which color to show
7. **Admin charts** → call the stats endpoint and map the data to chart widgets